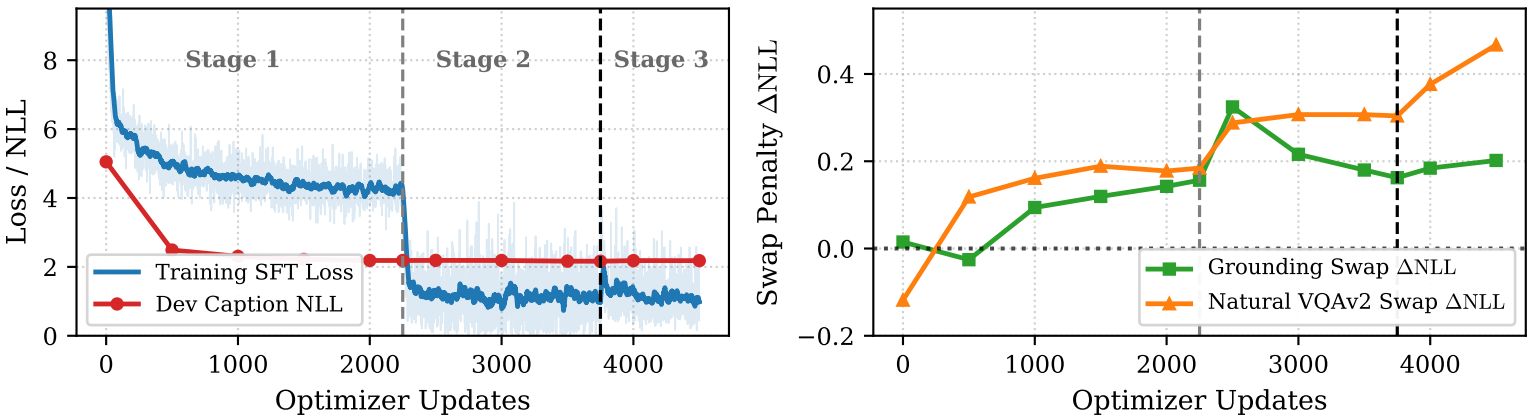

(a) Curriculum SFT Loss & Dev NLL
Stage 1
Stage 2
Stage 3
Loss / NLL
Training SFT Loss
Dev Caption NLL
Optimizer Updates
(b) Dev Image-Swap Penalty ΔNLL
Swap Penalty ΔNLL
Grounding Swap ΔNLL
Natural VQAv2 Swap ΔNLL
Optimizer Updates